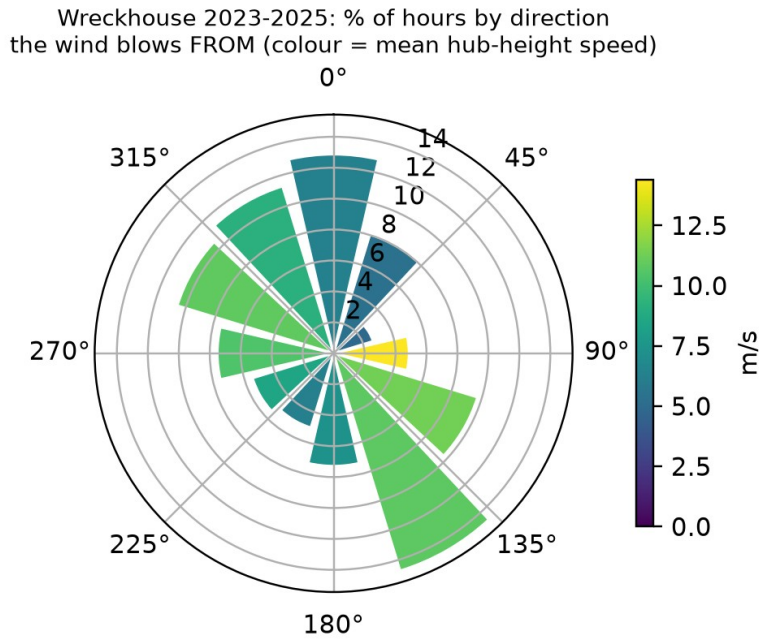
WakeWise NL

Wake-aware wind-turbine siting for Newfoundland's west coast, solved with QAOA in Qiskit and run on IBM Quantum System One in Quebec

Team AI Entanglers

Mostafa Wasib · Mahmoud Al-Moghrabi

Qiskit Fall Fest 2026 · Memorial University



Real input: 26,264 hours of wind at Wreckhouse, NL, 2023 to 2025

WakeWise NL in one minute

154 GWh/yr

from the recommended 9-turbine layout: the electricity of ~9,100 NL homes

~340x

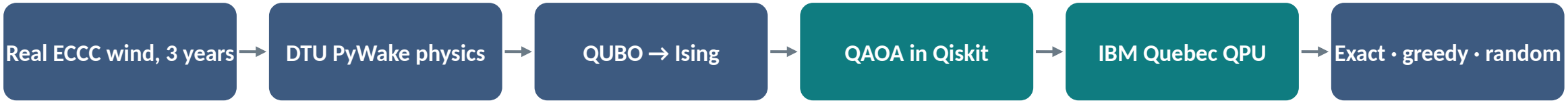
more often than random guessing, QAOA returns the optimal layout (12 qubits, noiseless simulation)

17%

of samples from IBM's Quebec machine beat the greedy planner on 2023's wind (random guessing: 2%)

2 of 3 years

the recommended layout is the best possible; in the third it gives up just 1 MWh/yr



Teal = quantum steps. No quantum advantage is claimed: at this size we can check every answer exactly.

Strong NL wind meets a proposed hydrogen farm

Wreckhouse, southwest Newfoundland

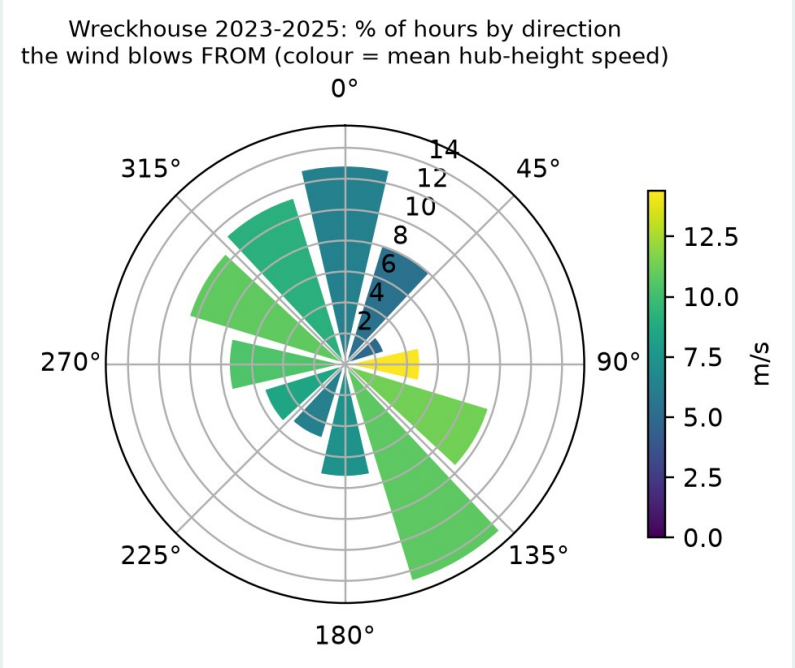
Gales here once blew trains off the tracks. Over 2023 to 2025 the wind at a 110 m hub averaged 9.0 m/s: a 4 MW turbine would run at a 50% capacity factor.

Project Nujio'qonik, next door

150+ turbines proposed on the Port au Port Peninsula and in the Codroy Valley to make hydrogen in Stephenville. Released from provincial environmental assessment, with conditions, in April 2024.

MUN's CREATE-SWIFT program asks for this

Benchmark wind-energy optimization: classical versus quantum or hybrid methods, on accuracy, cost, scalability and real-world relevance.



Bimodal wind: south-easterly gales and north-westerlies. Turbines lined up on that axis shade each other most.

Sources: ECCC hourly climate data (Wreckhouse, ID 8404343); NL Government EA bulletin, 9 Apr 2024; CBC News; CREATE-SWIFT talk, Qiskit Fall Fest 2026.

Wakes waste wind, so siting is a real decision

17,600 MWh/yr

from one 4 MW turbine standing alone at Wreckhouse (2023-2025 wind, DTU PyWake)

up to 2.4%

of that lost to one badly placed neighbour's wake; a dense 12-turbine block loses 4.2%

32% of hours

at full power, when a wake costs almost nothing: why wakes here are milder than at an average site

Decision and intended use

Which candidate sites get a turbine (bit = 1) or not (bit = 0). Output: the best layouts, the energy at stake, and whether the choice holds up year to year; for early-stage screening, not construction advice.

Sustainability benefit

More clean electricity, or green hydrogen, from the same land, steel and concrete; or the same energy from fewer turbines and less disturbed ground. The prototype shows the method; real benefit needs site-specific terrain, turbines and costs.

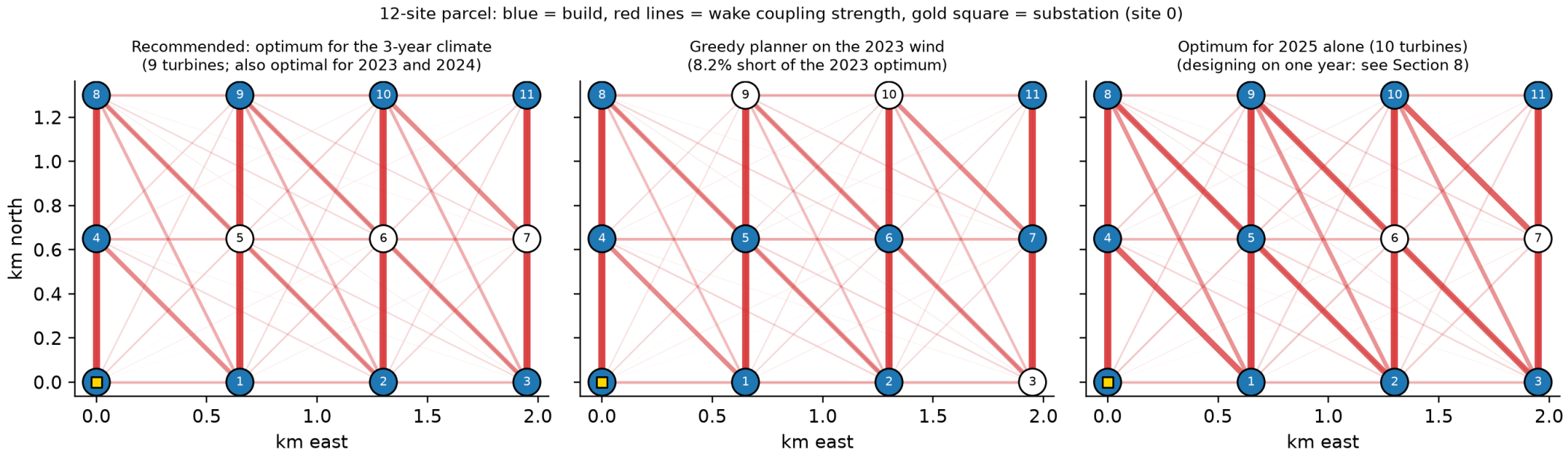
Inputs, output and scope

Input	Value used	Status
Hourly wind direction and speed	Wreckhouse, ECCC, 2023 to 2025 (26,264 hours)	**Observed**
Hub-height wind	10 m to 110 m with power-law shear, alpha = 0.14	Assumption
Wind-farm physics	DTU PyWake: Jensen wake model; checked with full-farm and Gaussian models	**Industry-standard model**
Turbine	4 MW, 130 m rotor, power and thrust curves from PyWake's generic model	Illustrative class
Parcel and costs	Flat grid, 650 m apart; cost = 88% of a turbine's output + 2% per 650 m of cable	Illustrative, sensitivity tested
Output	Build / no-build bitstring = one layout	
Scope	12 sites (12 qubits) on the 3-year climate and on each single year; 16 sites as a scale test; simulation, IBM noise model and real hardware	

Wake interactions make the choice combinatorial

$$f(x) = -\Sigma\,(E - c_i)\,x_i \;+\; \Sigma\,W_{ij}\,x_i\,x_j$$

E: one turbine's energy · c_i: annualized cost as energy · W_{ij}: energy the pair loses to each other's wakes (PyWake) · MWh/yr, lower is better



Greedy (add the best site until none helps) finds the 3-year optimum, but falls 2% to 8% short in single years. Every layout is checked exactly against all 4,096 options.

The wake physics is already an Ising model

$$x_i = (1 - z_i) / 2$$

$$H = \Sigma\, h_i Z_i + \Sigma\, J_{ij} Z_i Z_j$$

Turbine value becomes a Z field; the wake loss between two turbines becomes a ZZ coupling.

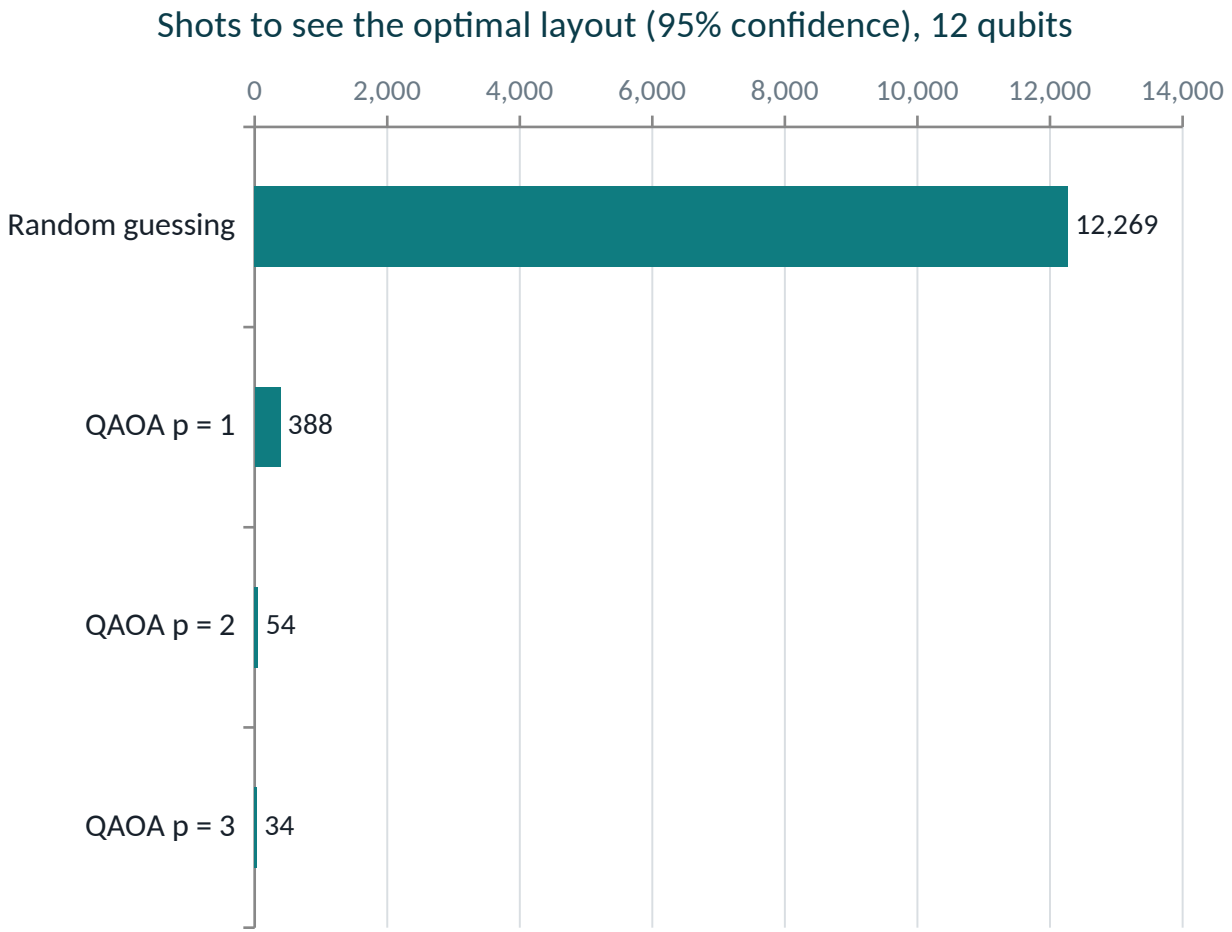
Why QAOA fits

- Natively quadratic and binary: no encoding overhead, no penalty terms
- NP-hard in general (contains maximum-weight independent set)
- CVaR training: optimize the best 25% of outcomes, because a planner needs the best layout
- Small enough to check exactly, so quality is measured, not assumed

1. **Map** *classical*
   ECCC wind → PyWake → QUBO → SparsePauliOp → qaoa_ansatz, depth p = 1 to 3
2. **Optimize** *classical*
   Transpile for IBM Heron (level 3); drop weak couplings to save two-qubit gates
3. **Execute** *quantum*
   Train angles on exact simulation (CVaR + COBYLA); sample layouts: ideal, noise model, ibm_quebec
4. **Post-process** *classical*
   Score every sampled layout with the full QUBO; compare with exact, greedy and random

QAOA finds the best layout ~340x more than chance

Noiseless simulation



8.4% vs 0.02%

of shots return the optimal layout (QAOA p = 3 vs random guessing)

34 vs 12,269

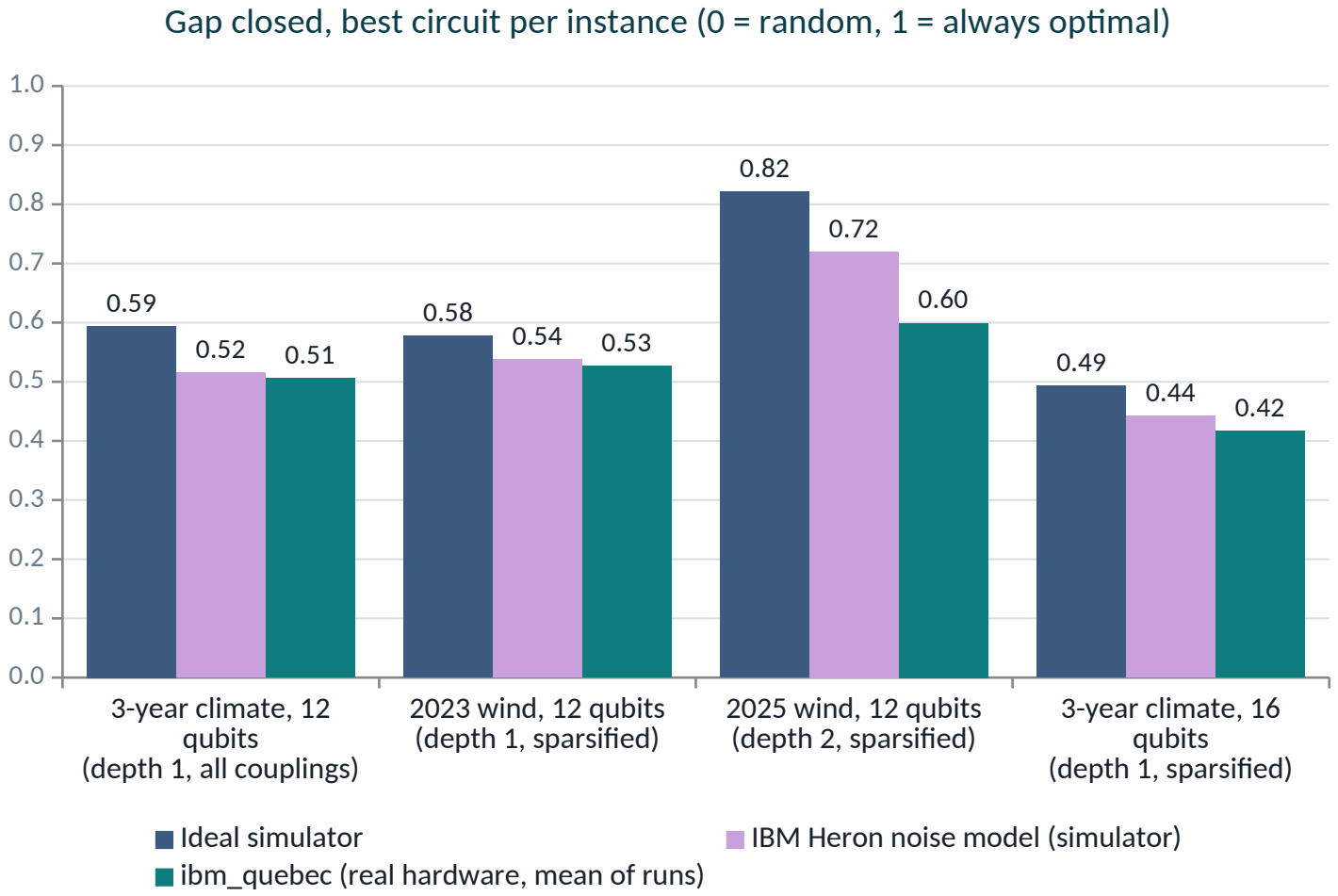
shots needed to see the optimum with 95% confidence

9% to 12%

of shots hit each single year's optimum, where greedy falls short; 13% to 59% of samples beat greedy

Honest caveat: a typical QAOA sample (gap closed 0.76) is below the best layout; QAOA's value is in its best few dozen shots.

On IBM's Quebec machine, the signal survives



Simulator + real hardware

ibm_quebec (IBM Quantum System One, PINQ², Bromont, Quebec), 7 Nov 2026; median CZ error 0.15%; 8 jobs, 56 s of QPU time.

3-year climate, 3 runs: the optimum in 0.47% to 0.60% of shots, 22x random.

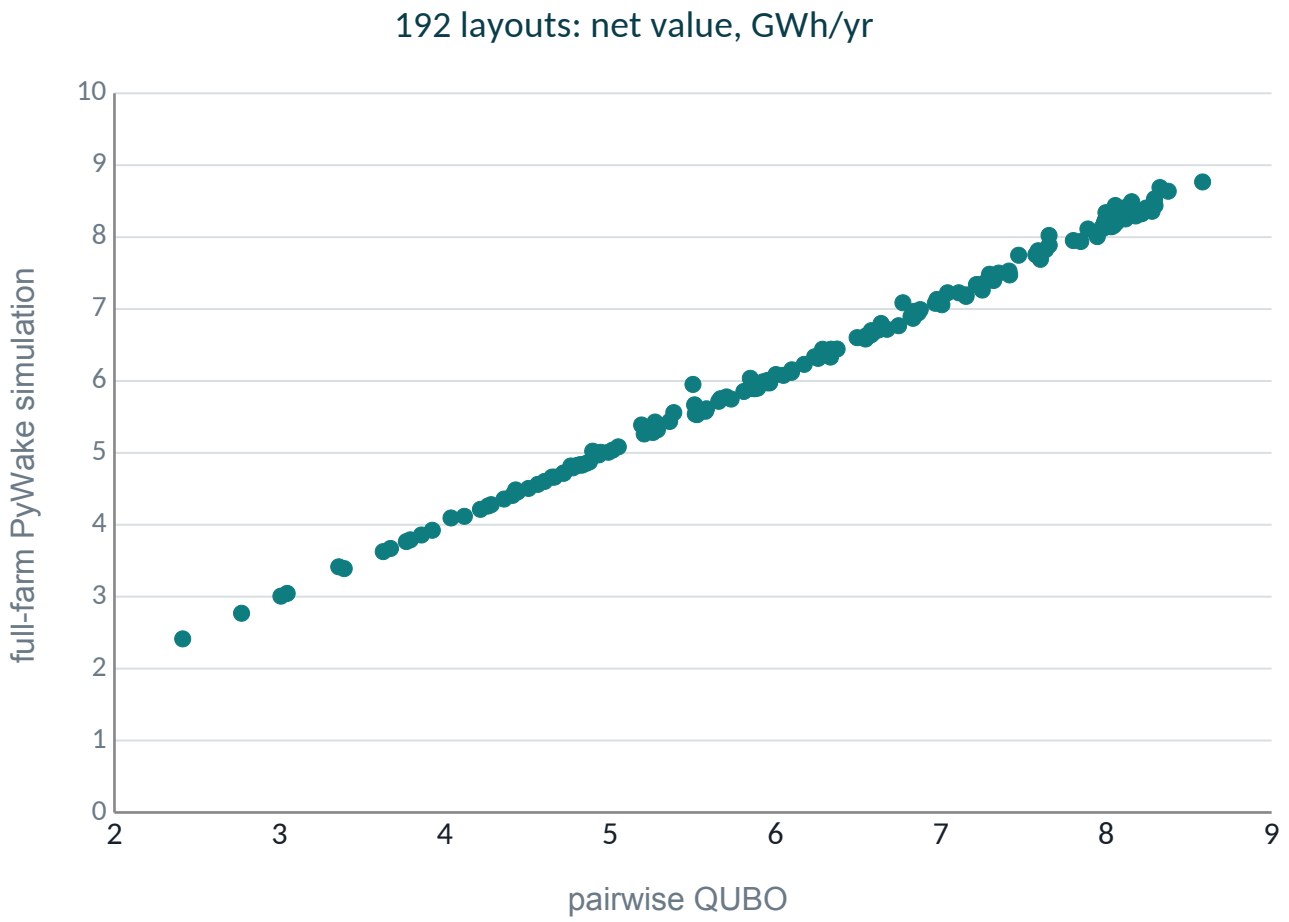
2023 wind, 3 runs: 17% of hardware samples beat the greedy planner (random guessing: 2%).

2025 wind, 1 run: the optimum 68x more often than random.

16 qubits: still clearly better than random (gap closed 0.42), but the optimum itself is rarely sampled: today's limit.

Control circuits (should be random) score -0.01: the gain is QAOA, not noise.

Our QUBO matches full wind-farm simulations



0.993

rank correlation with full-farm PyWake (all wakes at once)

0.992

rank correlation with a different, Gaussian wake model

#1 and #1

rank of the QUBO's optimal layout under those two simulations

The QUBO adds wake losses pair by pair; a full simulation lets every wake act at once. They agree, so optimizing the QUBO optimizes the farm.

The recommended layout holds up in every year

Energy given up versus each year's best layout (MWh/yr): lower is better

Layout designed on	2023	2024	2025	Average
3-year climate (recommended; also the 2023 and 2024 optimum)	**0**	**0**	**1**	**0**
2025 wind only	357	416	0	258

Designing on one year is risky: the 2025-only layout gives up hundreds of MWh/yr in calmer years. The long-term layout is the best possible in 2023 and 2024 and nearly so in 2025.

2% to 8%

greedy shortfall on single years

9% to 12%

of QAOA shots (p = 3, noiseless) hit each year's optimum

17%

of real-hardware samples beat greedy on the 2023 wind

What the evidence supports, and what it does not

Supports

- Siting is combinatorial on real NL wind, and the pairwise QUBO matches full wind-farm simulations
- The recommended layout is robust across three real wind years; single-year designs are not
- QAOA biases sampling toward the best layout (~340x random at p = 3) and improves with depth
- The signal survives real hardware, repeatably: 22x random over 3 runs; 17% of 2023 samples beat greedy

Does not support

- Quantum advantage: a laptop checks all 4,096 layouts in milliseconds
- Beating greedy on the long-term climate: greedy is already optimal there
- Real-world energy figures: flat idealized parcel, illustrative costs
- Scaling on today's hardware: at 16 qubits the optimum is rarely sampled

Limitations and next steps

Limitations and resources

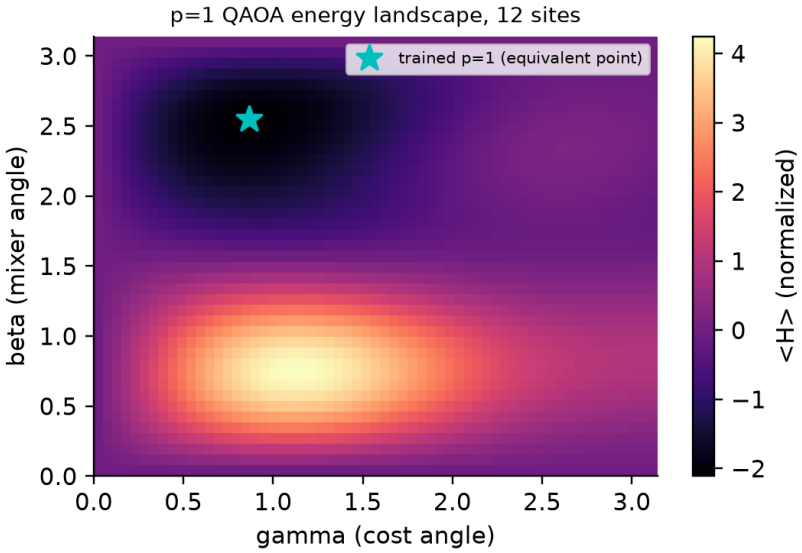
- Model: flat terrain, 10 m data extrapolated to hub height, a generic turbine class, three wind years
- Quantum: depth $p \leq 3$ trained in simulation; heavy-hex SWAPs keep hardware circuits shallow; 16 qubits is today's limit here
- Resources: the notebook runs in about 15 minutes on a laptop; hardware used 8 jobs and 56 s of QPU time

Next steps

1. Real Codroy Valley parcel with terrain-aware wind
2. Spacing and exclusion zones (wetlands, caribou habitat) as extra QUBO terms
3. Constraint-preserving mixers and warm starts
4. Scale past exact search; compare with simulated annealing and MILP
5. Partner route: MUN CREATE-SWIFT, with Mitacs Accelerate funding

More clean energy from the same wind

- Three years of real Wreckhouse wind + DTU PyWake physics give a QUBO that matches full wind-farm simulations
- The recommended layout is robust across real wind years
- QAOA in Qiskit finds it ~340x more often than chance; on IBM's Quebec machine it repeatably beats random guessing, and 17% of its samples beat greedy on 2023's wind



p = 1 QAOA energy landscape for the 12-site parcel; the star marks the trained angles.

github.com/mrw-soumik/Qiskit-AI-Entanglers-

Thank you. Questions?

Credits: QAOA workflow adapted from Tinh Bui's workshop notebook; IBM Qiskit 101 notebook and Fall Fest hardware example; DTU PyWake; ECCC data (Open Government Licence). Full credits in the notebook.